# Propositional Logic Worksheet: Translating Sentences into Symbols

**This document is provided for reference to lecture content, you do not need to submit this problem set.**

**Notation:** $\sim$ (not), $\wedge$ (and), $\vee$ (or), $\rightarrow$ (if…then), $\leftrightarrow$ (if and only if)

### A note on the conditional

A statement of the form P $\rightarrow$ Q is called a **conditional**. P is called the **antecedent** (the "if" part) and Q is called the **consequent** (the "then" part).

The conditional is only *false* when a true antecedent leads to a false consequent — that is, when the statement makes a promise it doesn't keep. In every other case it is *true*. In particular, **whenever the antecedent is false, the conditional as a whole is true**, regardless of the truth value of the consequent. (Think of it this way: if the antecedent never happens, the conditional's promise is never broken, so it counts as true.)

For each problem below, write your symbolic translation in the blank top cell of the right-hand column of the truth table, then fill in every row below it: the settings of the propositions on the left, and the resulting truth value of your formula on the right.

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### Worked Example 1

**Statement:** It is raining and it is cold. **Propositions:** R = it is raining, C = it is cold

**Symbolic form:** R $\wedge$ C

**Truth table:**

| R | C | **R $\wedge$ C** |
|---|---|---|
| T | T | T |
| T | F | F |
| F | T | F |
| F | F | F |

## Worked Example 2

**Statement:** If the alarm sounds, then the building will be evacuated. **Propositions:** A = the alarm sounds, E = the building is evacuated

**Symbolic form:** $A \to E$

Here A (the alarm sounds) is the antecedent and E (the building is evacuated) is the consequent.

**Truth table:**

| A | E | $\mathbf{A \to E}$ |
|---|---|---|
| T | T | T |
| T | F | F |
| F | T | T |
| F | F | T |

Notice the last two rows: whenever the antecedent A is false, the conditional is true no matter what E is.

## Exercises

### Part 1 — Single Connective

**1.** It is raining and it is cold. *Propositions:* R = it is raining, C = it is cold

**Truth table:**

| R | C | |
|---|---|---|
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**2.** Either Maria will call or she will text. *Propositions:* M = Maria calls, T = Maria texts

**Truth table:**

| M | T | |
|---|---|---|
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**3.** If the alarm sounds, then the building will be evacuated. *Propositions:* A = the alarm sounds, E = the building is evacuated

**Truth table:**

| A | E | |
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**4.** The store is not open. *Propositions:* O = the store is open

**Truth table:**

| O | |
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**5.** John is tired, but he is not giving up. *Propositions:* T = John is tired, G = John is giving up

**Truth table:**

| T | G | |
|---|---|---|
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## Part 2 — Two Connectives

**6.** If it snows and the roads are icy, then school will be closed. *Propositions:* S = it snows, I = the roads are icy, C = school is closed

**Truth table:**

| S | I | C | |
|---|---|---|---|
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**7.** Either the printer is out of paper or it is jammed, and it needs attention. *Propositions:* P = printer is out of paper, J = printer is jammed, A = it needs attention

**Truth table:**

| P | J | A | |
|---|---|---|---|
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**8.** If Sam is not home, then Lee will feed the cat. *Propositions:* H = Sam is home, F = Lee feeds the cat

**Truth table:**

| H | F | |
|---|---|---|
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**9.** The flight is delayed or cancelled, but the passengers have not been notified. *Propositions:* D = the flight is delayed, C = the flight is cancelled, N = the passengers have been notified

**Truth table:**

| D | C | N | |
|---|---|---|---|
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**10.** If the test is passed, then the license is issued and the fee is refunded. *Propositions:* T = the test is passed, L = the license is issued, F = the fee is refunded

**Truth table:**

| T | L | F | |
|---|---|---|---|
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### Part 3 — Negation with Conjunction/Disjunction

**11.** It is neither raining nor snowing. *Propositions:* R = it is raining, S = it is snowing

**Truth table:**

| R | S | |
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**12.** Not both the door and the window are locked. *Propositions:* D = the door is locked, W = the window is locked

**Truth table:**

| D | W | |
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**13.** Either the battery is dead or the switch is broken, and it is not the fuse. *Propositions:* B = the battery is dead, S = the switch is broken, F = it is the fuse

**Truth table:**

| B | S | F | |
|---|---|---|---|
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**14.** The report is late, and it was not reviewed. *Propositions:* L = the report is late, R = the report was reviewed

**Truth table:**

| L | R | |
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### Part 4 — Nested Conditionals and Mixed Structure

**15.** If Ana studies, then if she rests well, she will pass the exam. *Propositions:* S = Ana studies, R = Ana rests well, P = Ana passes the exam

**Truth table:**

| S | R | P | |
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**16.** If the power fails or the server crashes, then the backup system starts. *Propositions:* P = the power fails, S = the server crashes, B = the backup system starts

**Truth table:**

| P | S | B | |
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**17.** If the coupon is valid, then the price is reduced or the shipping is free. *Propositions:* V = the coupon is valid, P = the price is reduced, F = the shipping is free

**Truth table:**

| V | P | F | |
|---|---|---|---|
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**18.** If it is not sunny and not warm, then the picnic is cancelled. *Propositions:* S = it is sunny, W = it is warm, C = the picnic is cancelled

**Truth table:**

| S | W | C | |
|---|---|---|---|
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## Part 5 — Biconditional

**19.** The light turns on if and only if the switch is up and the circuit is not broken. *Propositions:* L = the light turns on, S = the switch is up, C = the circuit is broken

**Truth table:**

| L | S | C | |
|---|---|---|---|
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**20.** You may enter the contest if and only if you are a member or you have paid the fee. *Propositions:* E = you may enter the contest, M = you are a member, F = you have paid the fee

**Truth table:**

| E | M | F | |
|---|---|---|---|
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